**Bahria University, Islamabad Campus**

**Course Title**: [CSC 411] Artificial Intelligence
**Instructor**: Dr.Kashif Sultan
**Date**: 10th April, 2026
**Due Date:** 15th April 2026

# Assignment # 02

**(CLO3)**

**Instructions:**

- Each student is expected to submit their own original work. Plagiarism or copying from other sources, including fellow students' assignments, is strictly prohibited.
- You are encouraged to explore various sources of information, including internet resources, books, and scholarly articles, to gather relevant information for your answers.
- Your answers should be written in your own words, demonstrating your understanding of the concepts presented in the scenarios. Avoid direct copying of text from external sources.
- Ensure that your answers are concise and to the point. Provide relevant examples or samples where necessary to support your explanations.
- If you refer to specific sources in your answers, be sure to provide appropriate citations or references.
- The viva of assignments will be conducted at the end of the course.

## Question:

In this assignment, you are required to use the already configured Docker-based Hadoop environment (as demonstrated in class) to perform advanced HDFS operations and analysis.
Show Docker installation or image pulling steps. However, your main focus should be on HDFS working, commands, and behavior analysis.

**Tasks Details:**

**Part 1: Hadoop Cluster Verification**

Verify that your Hadoop cluster (NameNode + DataNodes) is running properly and access the NameNode Web UI (localhost:9870 or 50070). Identify:

- Number of DataNodes
- Cluster status (Live/Dead nodes)
- Storage capacity

**Part 2: HDFS Directory & File Operations**

Perform the following operations inside HDFS:

- Create the following directory structure:
  - /student/<your_name>/data
- Create at least **two local files** (e.g., text files with sample data)
- Upload both files into HDFS
- Display file contents using HDFS commands
- List all files with detailed information

**Part 3: Replication Factor Experiment**

- Upload a file with default replication
- Change its replication factor (e.g., 1 → 2 or 3)
- Verify the change using:
  - hdfs dfs -stat %r <file>
- Explain:
  - What happens when replication factor increases?
  - How does it affect storage?

**Part 4: HDFS Block & Storage Analysis**

- Use command:

hdfs fsck <file> -files -blocks -locations

- Identify:
  - Number of blocks
  - Block locations (DataNodes)
- Explain how HDFS distributes data across nodes

**Part 5: Cluster Health Monitoring**

- Run:

hdfs dfsadmin -report

- Analyze:
  - Total capacity
  - Used vs remaining space

    - Status of each DataNode
- Take screenshot and briefly interpret the output

**Part 6: Fault Tolerance Observation (Conceptual + Practical)**

- Stop one DataNode container
- Observe:
    - Changes in cluster (UI or report)
    - Availability of files
- Restart the DataNode
- Explain:
    - How HDFS handles failures
    - Role of replication in fault tolerance

**Submission Requirements:**

- Submit a **well-formatted PDF report**
- Include:
    - Clearly labeled screenshots of each task
    - Commands used
    - Brief explanation (2–4 lines per step)

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